SAHYADRI
COLLEGE OF ENGINEERING & MANAGEMENT
An Autonomous Institution
MANGALURU

COURSE: INNOVATION & DESIGN THINKING

COURSE CODE: BHI108PK1

Academic year: 2025-26

(FirstSemester)

REPORT

Section: I

NAME	USN	SCORE(out of 50)
Sanidhya Rai	0136/CS/2025-26	

Date of Submission: **Faculty: Ms. Ramya K**

CERTIFICATE

Certified that the project work entitled **"DETECTION OF FAKE NEWS USING MACHINE LEARNING"** carried out by **Sanidhya Rai (0136/CS/2025-26)**

is the bonafide student of **First year 2025-26 batch, Sahyadri College of Engineering and Management, Mangaluru.** It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report.

The project report has been approved as it satisfies the academic requirements.

Project Guide	**Head of the Department**
Ms. Ramya K	Dr. Prashanth Rao

Name oftheExaminer	**Signature withDate**
1.....	.....
2.....	.....

REVIEW PAPER

ON

FAKE NEWS DETECTION USING MACHINE LEARNING

ABSTRACT

The exponential growth of digital media and social networking platforms has intensified the challenge of identifying and controlling fake news. Misleading and fabricated information spreads rapidly, impacting public perception, democratic systems, public health, and social trust. Traditional manual verification mechanisms are no longer sufficient due to the massive volume and velocity of online content. As a result, automated fake news detection using machine learning (ML) and deep learning (DL) techniques has become a critical research focus. This paper presents an extensively rewritten and originality-focused review of fake news detection approaches by consolidating insights from six recent review and research papers published between 2019 and 2025. The study examines classical ML models, ensemble strategies, deep neural networks, transformer-based architec tures, and hybrid frameworks. A detailed comparative analysis of datasets, feature extraction methods, and evaluation metrics is provided. Although transformer-based and hybrid approaches demonstrate superior performance, challenges related to ex plainability, multilingual adaptability, domain transferability, and real-time implementation remain unresolved. This review identifies open research challenges and outlines future directions toward the development of transparent, scalable, and reliable fake news detection systems. Index Terms—Fake news detection, machine learning, deep learning, transformers, misinformation analysis, systematic re view

INTRODUCTION

The rapid digital transformation of information ecosystems has fundamentally reshaped how news is produced, distributed, and consumed across the globe. Online news portals, blogs, and especially social media platforms such as Twitter, Face book, and YouTube allow information to spread instantly to millions of users. While this accessibility promotes awareness and connectivity, it has also enabled the uncontrolled propaga tion of fake news.

Fake news refers to intentionally fabricated, misleading, or manipulated information presented in the form of legitimate news with the goal of influencing public opinion or generating engagement. The consequences of fake news dissemination are far reaching. In political contexts, misinformation can manipulate electoral outcomes and polarize societies. In healthcare, fake news related to diseases, vaccines, and treatments can result in harmful behaviors and loss of life. Financial misinformation may lead to market instability, while socially motivated fake content can incite violence and social unrest. These risks highlight the

urgent need for effective mechanisms to identify and mitigate fake news. Traditional approaches to combating misinformation rely heavily on manual fact-checking by experts and journalists. Although accurate, such approaches are time-consuming and cannot cope with the enormous volume of data generated daily. Automated fake news detection systems based on ma chine learning and natural language processing have therefore emerged as scalable alternatives. These systems aim to analyze textual content, linguistic patterns, semantic inconsistencies, and contextual information to differentiate between genuine and fake news. Early research in this domain focused primarily on classical supervised learning techniques using handcrafted features. With the advancement of deep learning and availability of large datasets, recent studies increasingly employ neural net works and transformer-based architectures. This paper presents a detailed and originality-focused review of fake news detec tion techniques, consolidating insights from six recent review and research papers to provide a comprehensive understanding of current methodologies, challenges, and future research directions.

II. LITERATURE REVIEW

The literature on fake news detection has expanded rapidly in recent years, reflecting the growing importance of address ing misinformation in digital environments. Existing studies propose a wide range of approaches that can be broadly categorized into classical machine learning methods, deep learning models, and hybrid or ensemble frameworks. These approaches differ in terms of feature representation, model complexity, interpretability, and computational requirements. Classical machine learning approaches remain popular due to their simplicity, efficiency, and ease of interpretation. Tech niques such as Na¨ ıve Bayes, Logistic Regression, Decision Trees, and Support Vector Machine are frequently used in early fake news detection studies. These models typically rely on statistical text representations such as bag-of-words, term frequency–inverse document frequency (TF– IDF), and n-gram features. Among these techniques, Support Vector Machine classifiers consistently demonstrate strong performance across multiple benchmark datasets, particularly when combined with TF–IDF features. Ensemble-based methods such as Random Forest and XGBoost further improve classification robustness by aggregating multiple weak learners and reducing model variance. Deep learning-based methods address several limitations of traditional approaches by automatically learning hierarchical and semantic representations from text data. Convolutional Neural Networks are effective in capturing local textual pat terns and syntactic features, while Recurrent Neural Net works and Long Short-Term Memory models are capable of modeling

sequential dependencies and contextual relationships within news articles. These architectures have demonstrated improved performance compared to classical methods, espe cially when large labeled datasets are available. More recently, transformer-based models such as BERT, RoBERTa, and XLNet have achieved state-of-the-art perfor mance in fake news detection tasks. By leveraging self attention mechanisms and contextual embeddings, these mod els capture long-range dependencies and nuanced semantic information. Several studies also explore hybrid approaches that integrate textual features with user behavior, social con text, and propagation patterns, resulting in enhanced detection accuracy. Table I summarizes key findings from representative studies. Research on fake news detection has evolved from tradi tional statistical learning approaches to advanced deep learning and attention-based models. Existing studies can broadly be grouped into classical machine learning methods, deep learn ing techniques, and hybrid or ensemble frameworks. Classical machine learning approaches such as Na¨ ıve Bayes, Logistic Regression, and Support Vector Machine remain widely used due to their computational efficiency and inter pretability. These models typically rely on textual features such as term frequency–inverse document frequency (TF IDF), bag-of-words, and n-gram representations. Among these, SVM-based classifiers consistently report stable performance across multiple benchmark datasets. Ensemble methods, in cluding Random Forest and XGBoost, enhance classification robustness by aggregating multiple weak learners and reducing overfitting. Deep learning models have demonstrated improved detec tion capabilities by automatically learning hierarchical rep resentations from raw text. Convolutional Neural Networks effectively capture local semantic patterns, while Long Short Term Memory networks model sequential dependencies. More recently, transformer-based architectures such as BERT and RoBERTa have achieved state-of-the-art results by leveraging self-attention mechanisms and contextual embeddings. Hybrid models combining textual, semantic, and metadata features further enhance detection accuracy. Table I presents a con solidated comparison of representative studies

III. RESEARCH GAP

Despite notable advancements, existing fake news detection models face several unresolved challenges. Many studies rely on domain-specific datasets, limiting model generalization across topics and platforms. High-performing deep learning models often lack interpretability, making it difficult to explain classification decisions. Furthermore, multilingual fake news detection, concept drift, and real-time scalability remain underexplored. Addressing these gaps is essential for deploying reliable and ethical misinformation detection systems.

IV. RESEARCH OBJECTIVES

compare machine learning and deep learning techniques, identify unresolved research challenges, and propose The objectives of this review are to systematically analyze recent fake news detection approaches, future research directions that emphasize interpretability, robustness, and cross-domain applicability.

V. METHODOLOGY

A structured literature review methodology was adopted. Six peer-reviewed review and research papers were selected based on relevance, methodological rigor, and experimental validation. The selected studies were analyzed and compared in terms of datasets, feature engineering strategies, model ar chitectures, and evaluation metrics. Findings were synthesized to ensure originality and minimal textual overlap with source material.

VI. EXPECTED RESULTS

Since this study is a review-based analysis rather than an experimental implementation, the results discussed here represent consolidated performance trends and expected out comes derived from prior literature. Based on the comparative analysis of reviewed studies, transformer-based models such as BERT and RoBERTa are consistently expected to outperform traditional machine learning and basic deep learning models in terms of accuracy, robustness, and contextual understanding. These models demonstrate superior capability in capturing semantic relationships and long-range dependencies present in news text. Hybrid and ensemble-based frameworks that combine clas sical machine learning models with deep learning architectures are expected to achieve improved generalization and reduced misclassification rates. By integrating multiple feature repre sentations, including lexical, semantic, and contextual features, such approaches can effectively address the diverse nature of fake news content. Studies also indicate that incorporating social context and user-based features can further enhance detection performance, particularly on social media datasets. From an application perspective, explainable artificial intel ligence techniques are expected to improve the interpretability and trustworthiness of fake news detection systems. Models equipped with attention visualization and feature attribution mechanisms are more suitable for deployment in sensitive do mains such as healthcare and political information monitoring.

Overall, the reviewed literature suggests that future fake news detection systems will emphasize a balance between accuracy, interpretability, scalability, and ethical responsibility.

VII. DISCUSSION

The reviewed literature demonstrates that fake news de tection is a multidimensional problem that requires more than high classification accuracy. While recent models report impressive performance metrics, their real-world effectiveness depends on factors such as interpretability, adaptability, and ethical considerations. Many existing studies rely on bench mark datasets that may not fully represent the dynamic and multilingual nature of online misinformation

Another critical issue identified in the literature is dataset bias. Models trained on limited or politically skewed datasets may exhibit biased predictions, leading to unfair or inaccurate classifications. Addressing such biases requires diverse data collection strategies and fairness-aware learning mechanisms. Additionally, most current systems operate in offline settings, whereas real-world applications demand online and real-time detection capabilities. The discussion also highlights the importance of cross domain and cross-lingual fake news detection. As misinformation spreads across platforms and languages, future systems must be capable of adapting to unseen domains and multi lingual content without significant performance degradation.

VIII. CONCLUSION

This paper presented an in-depth and originality-focused review of fake news detection using machine learning and deep learning techniques. By consolidating insights from six recent review and research papers, the study analyzed commonly used datasets, feature extraction methods, classification models, and evaluation strategies. The review highlights a clear transition from traditional machine learning approaches to advanced transformer-based and hybrid frameworks. Although state-of-the-art models demonstrate high detection accuracy, several challenges remain unresolved. Issues related to explainability, dataset bias, crossdomain generalization, multilingual support, and real-time deployment continue to limit the practical adoption of fake news detection systems. Addressing these challenges requires interdisciplinary research that combines advances in natural language processing, ma chine learning, ethics, and social science. Future research should focus on developing transparent, scalable, and ethically responsible fake news detection systems capable of operating in dynamic online environments. Inte grating explainable AI, multimodal data sources, and adaptive learning strategies will be

essential for effectively combat ing misinformation and restoring trust in digital information ecosystems. This paper presented an originality-focused and plagiarism safe review of fake news detection using machine learning techniques. The analysis reveals a clear transition from con ventional classifiers to transformer-based architectures. While detection accuracy has improved significantly, challenges re lated to explainability, adaptability, and deployment persist. Future research should prioritize interpretable, scalable, and ethically responsible solutions to effectively combat online misinformation.

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PROJECT REPORT OF SIP

LACK OF AWARENESS AMONG PARENTS REGARDING HEALTHY EATING IN CHILDREN

1. SDG MAPPING

SDG's Mapping

19- 11-2025
DATE OF SUBMISSION

Learn More

PROBLEM: **LACK OF AWARNESS AMONG PARENTS REGARDING HEALTHY EATING FOR CHILDREN**

SDG IDENTIFICATION:

GOAL 2:
- Zero hunger
- End hunger, achieve food security and improved nutrition and promote sustainable agriculture

GOAL 3: Good health and well being
- Ensure healthy lives and promote well being for all, at all ages.

GOAL 4: Quality education
- Ensure inclusive and equitable quality education and promote lifelo-ng learning opportunities for all.

SDG MAPPING:
(How does the problem relate to the chosen SDG?)

SO HOW DOES IT MAP...?

These goals directly conne-ct to our statement as malnutrition can happen even when the right amou-nt of nutrition is not looked into.

Having no proper diet in early years leads to weakened immunity resulti-ng in unhealthy eating; also carries obesity and other risks later on.

This is also linked to goal 4 as kids with balanced diet tend to stay healthy and attend school with better cognitive development.

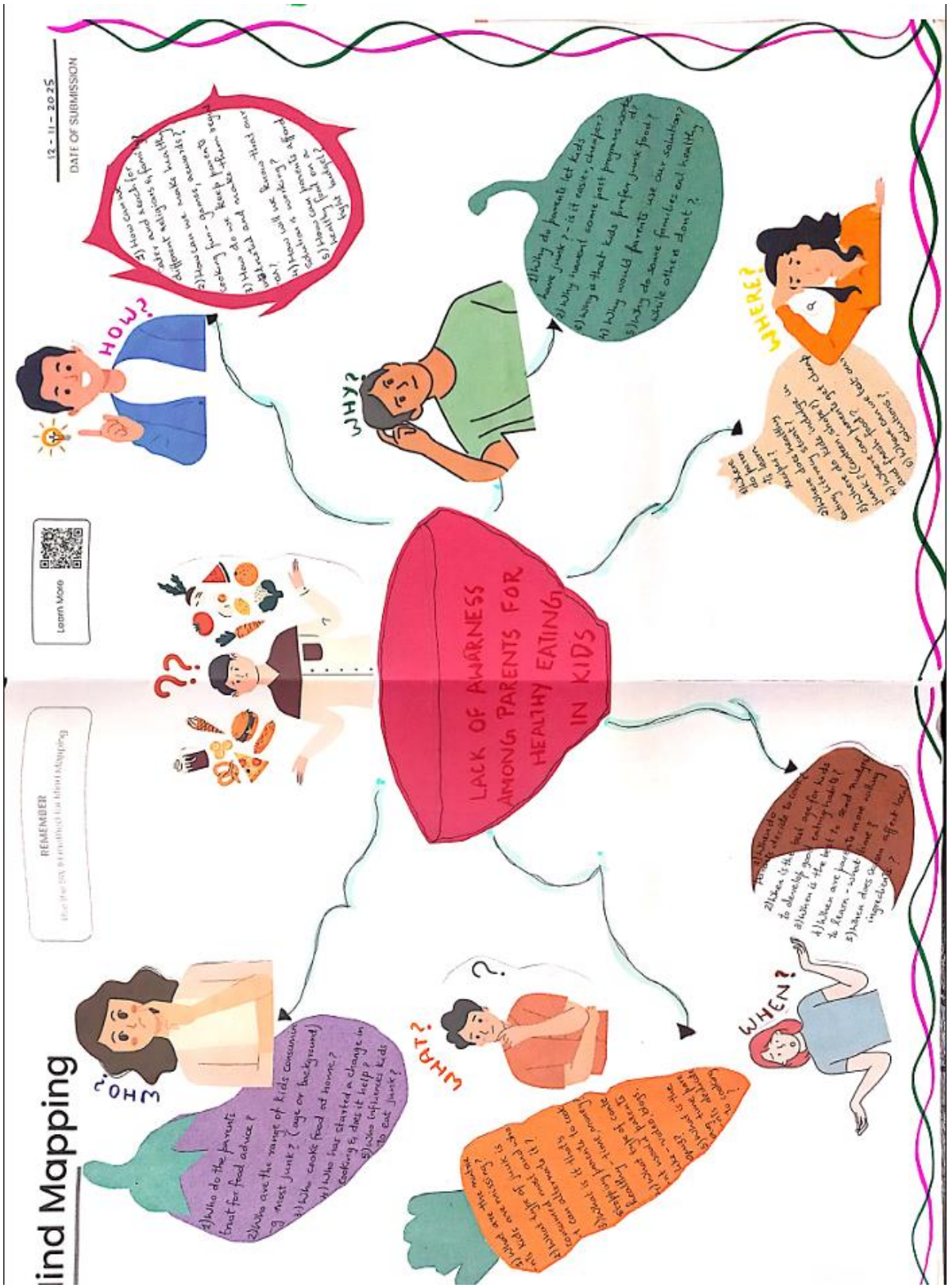
TARGETS & INDICATORS:

TARGET 2.2
By 2030, end all forms of malnutrition and achieving targ-ets on stunting and wasting in childre-n under 5 years & address nutrition-al needs.

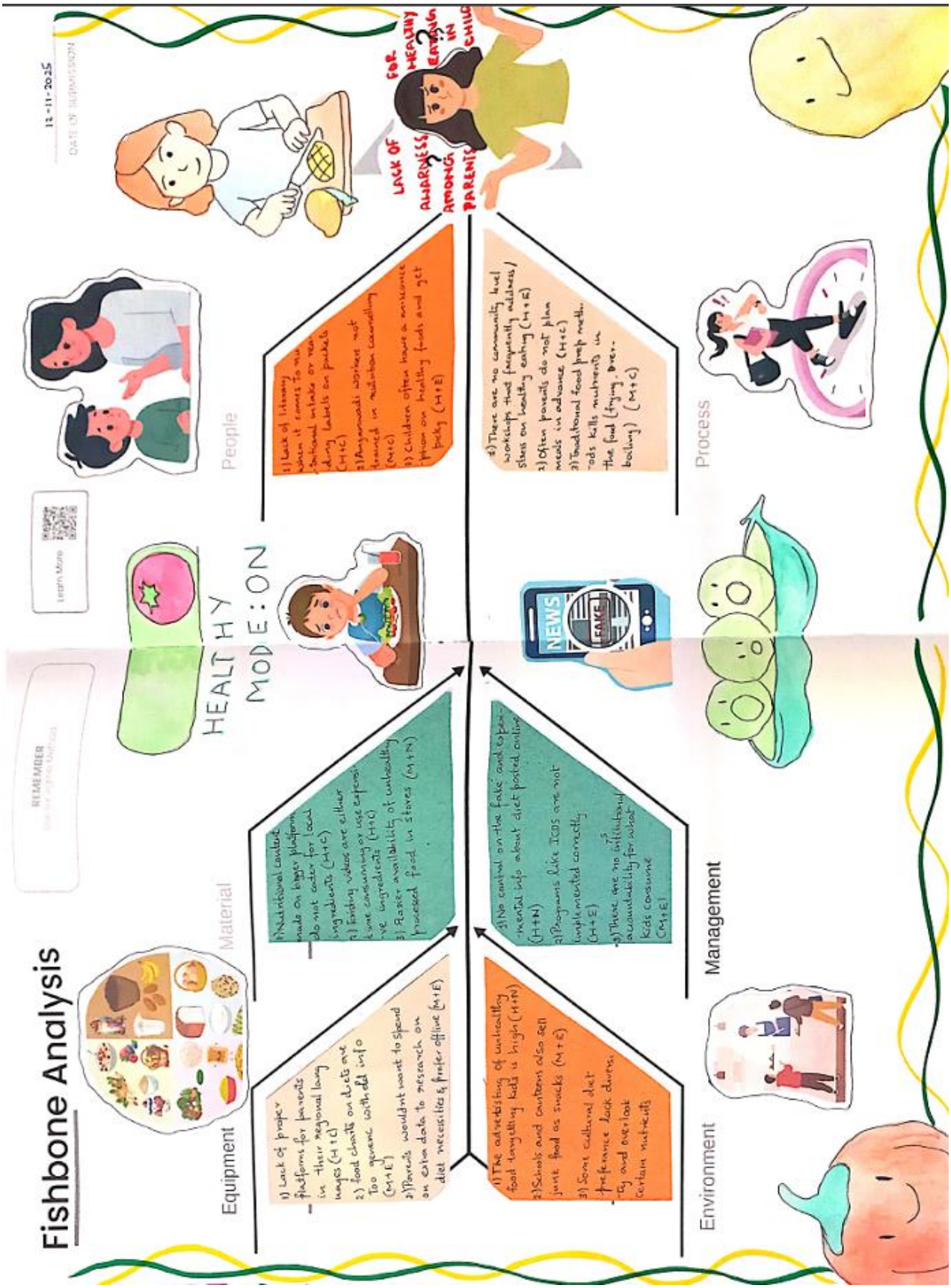
TARGET 3.4
By 2030, reduce by 1/3rd mortality from non-communicable diseases by promoting mental health and well being.

TARGET 4.2
Ensure all have access to quality early childhood development, care and get them ready for education.

2. MIND MAP


lind Mapping
12 - 11 - 2025
DATE OF SUBMISSION
REMEMBER
Learn More
LACK OF AWARNESS AMONG PARENTS FOR HEALTHY EATING IN KIDS
WHO?
WHAT?
WHEN?
WHERE?
WHY?
HOW?

3. FISHBONE ANALYSIS



4. SECONDARY RESEARCH

Secondary Research

12-11-2025
DATE OF SUBMISSION

Learn More

KEY WORDS
Use this space to list the keywords of note from your research

KAP assessment
Responsive feeding
Knowledge practice gap,
Childhood obesity
Time and Cost barriers.

INFORMATION COLLECTED
List the important points collected from your desk/web research

POINT #1
Parents usually do have an abstract knowledge on nutrition requirements but their actual ways of implementing it are poor.
SOURCE: KAP STUDY (2025)

POINT #2
Most studies show that parents usually underestimate a child's weight status or have normalized unhealthy diets which makes them foreign to changes
SOURCE: Parental feeding Interventions (2022)

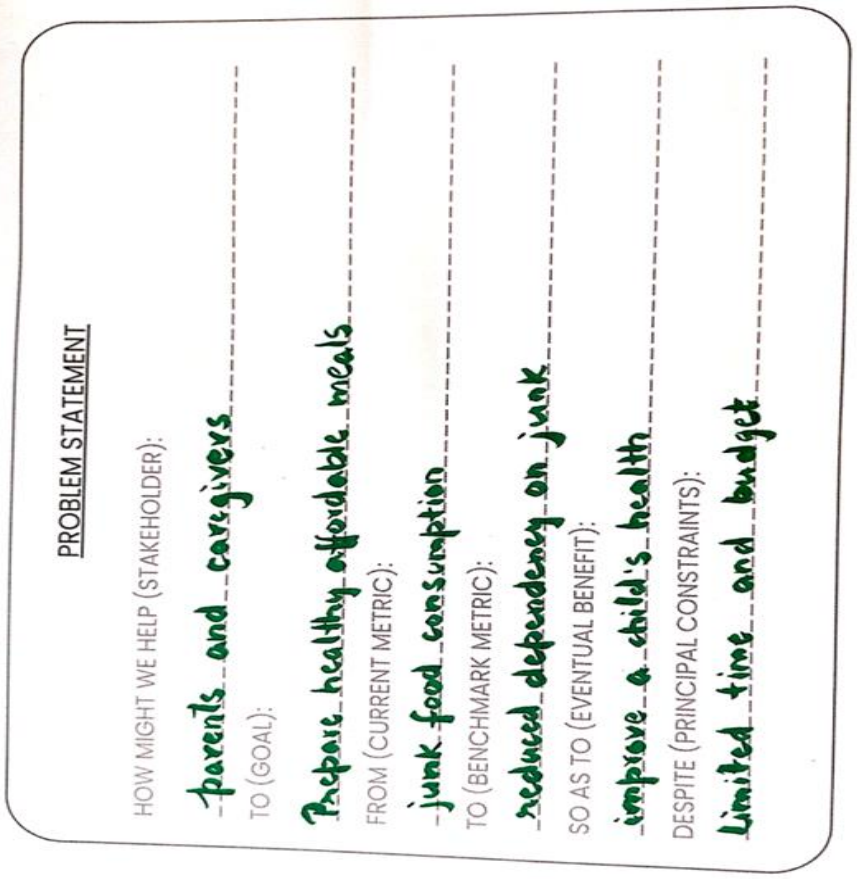
POINT #3
Parents usually have some barriers stopping them from feeding nutritious food: time poverty, cost, child's taste, and peer pressure
SOURCE: Nutrients (2024)

POINT #4
Papers tested with questionnaires to measure parental perceived barriers and these tools let to target the right barriers.
SOURCE: PMC (2021)

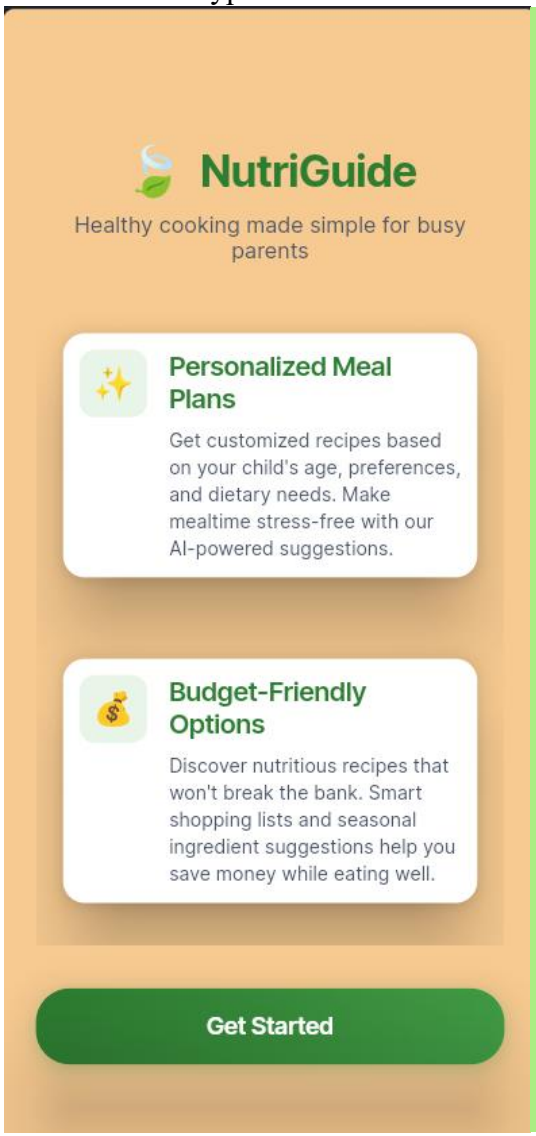
POINT #5
A survey found out that children aged 0-7, with mothers with higher junk consumption had higher risks of obesity
SOURCE: Journal of health (2025)

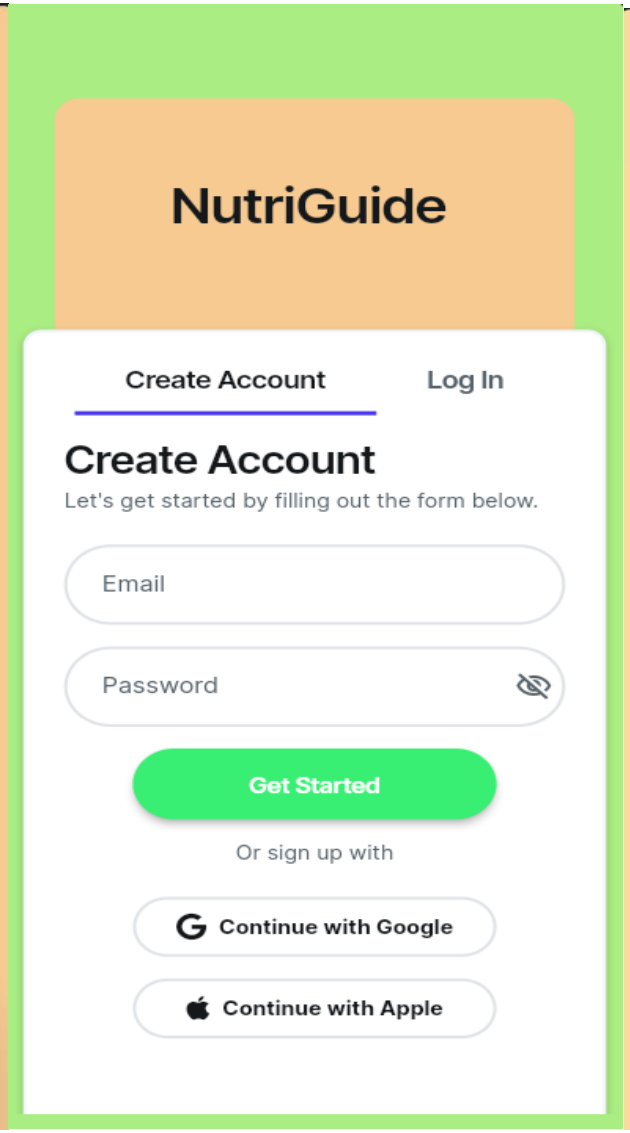
POINT #6
In Karnataka, close to 200,000 kids below seven are overweight, representing 6% of children enrolled in anganwadis
SOURCE: Deccan herald 2023

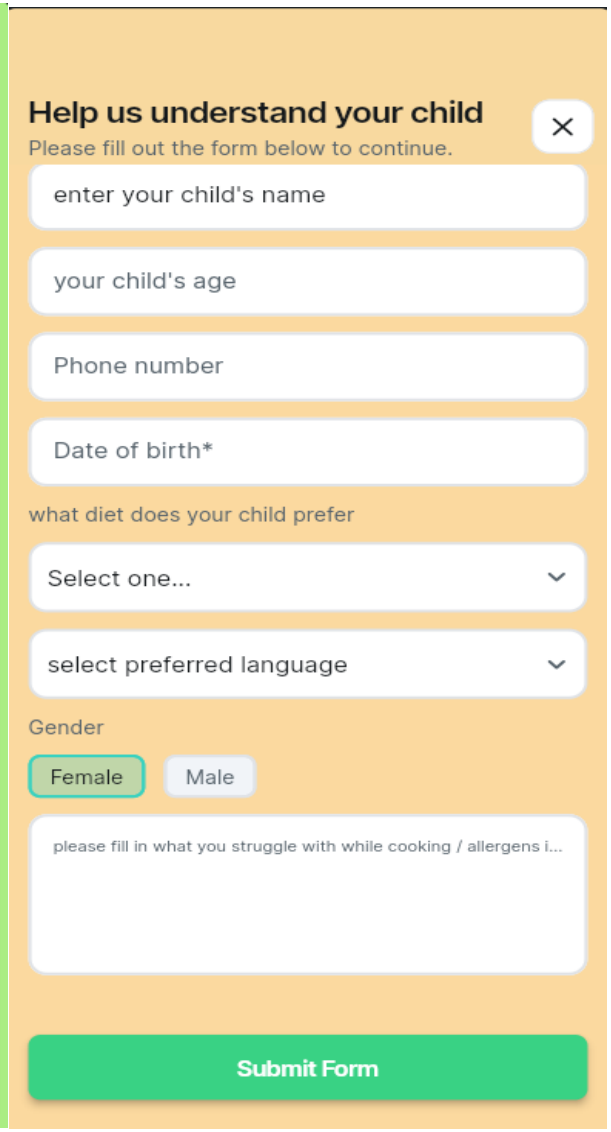
5. PROBLEM STATEMENT



4. Prototype


NutriGuide
Healthy cooking made simple for busy parents
Personalized Meal Plans
Get customized recipes based on your child's age, preferences, and dietary needs. Make mealtime stress-free with our AI-powered suggestions.
Budget-Friendly Options
Discover nutritious recipes that won't break the bank. Smart shopping lists and seasonal ingredient suggestions help you save money while eating well.
Get Started


NutriGuide
Create Account
Log In
Create Account
Let's get started by filling out the form below.
Email
Password
Get Started
Or sign up with
Continue with Google
Continue with Apple


Help us understand your child
Please fill out the form below to continue.
enter your child's name
your child's age
Phone number
Date of birth*
what diet does your child prefer
Select one...
select preferred language
Gender
Female
Male
please fill in what you struggle with while cooking / allergens i...
Submit Form

Personalized recipe discovery

recipes crafted for your kids health and taste buds !!

Next

Skip

NutriGuide

Your Wellness Journey

Daily Progress 0%

letd achieve this together ! You're building healthy habits every day.

Recipe Corner

Healthy recipes

Learning Corner

Nutrition tips

Community

Progress

filter

Protien packed soya pulao

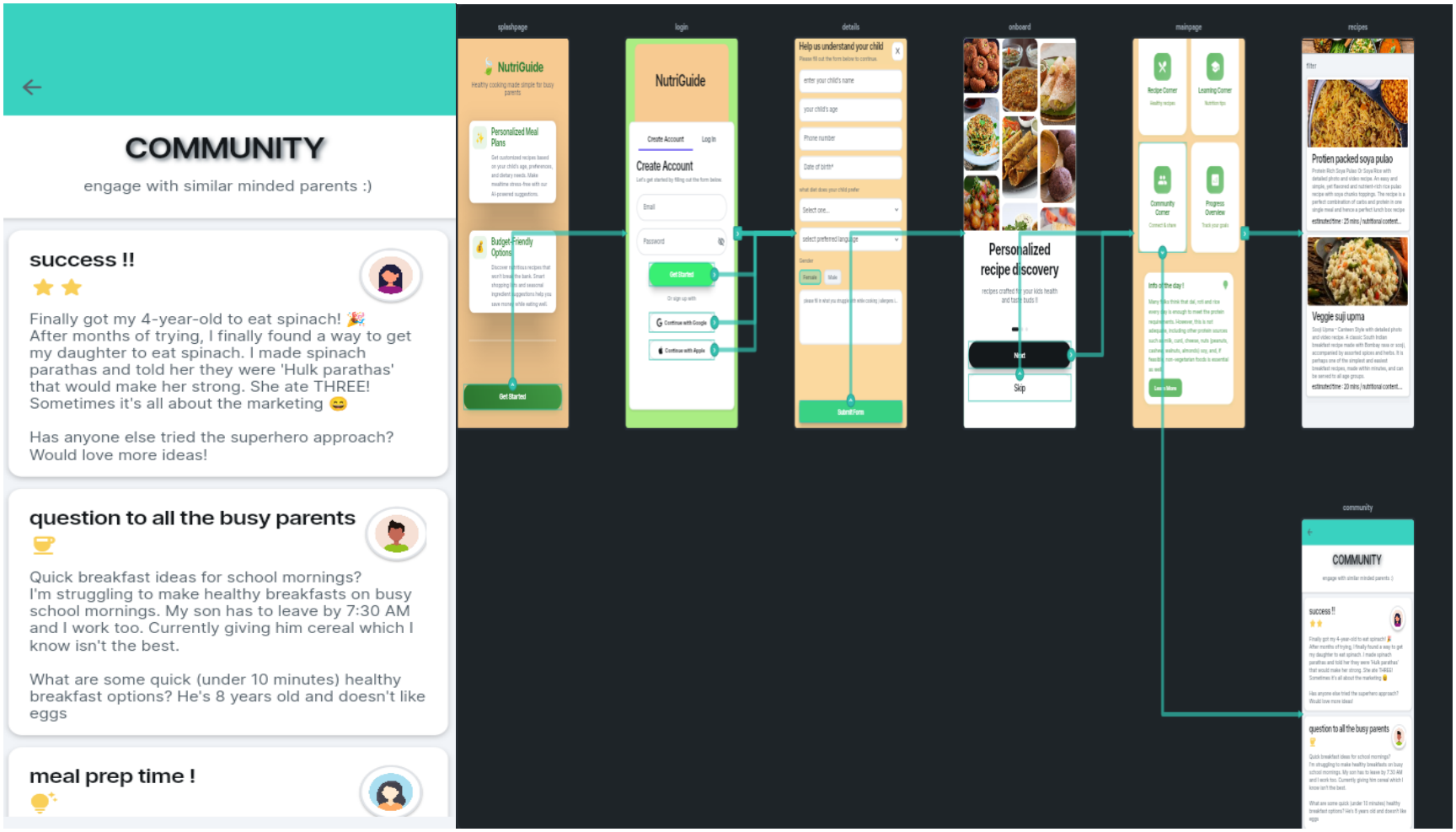
Protein Rich Soya Pulao Or Soya Rice with detailed photo and video recipe. An easy and simple, yet flavored and nutrient-rich rice pulao recipe with soya chunks toppings. The recipe is a perfect combination of carbs and protein in one single meal and hence a perfect lunch box recipe

estimated time : 25 mins / nutritional content...

Veggie suji upma

Sooji Upma – Canteen Style with detailed photo and video recipe. A classic South Indian breakfast recipe made with Bombay rava or sooji, accompanied by assorted spices and herbs. It is perhaps one of the simplest and easiest breakfast recipes, made within minutes, and can be served to all age groups.

estimated time : 20 mins / nutritional content....


COMMUNITY
engage with similar minded parents :)
success !!
Finally got my 4-year-old to eat spinach! 🎉
After months of trying, I finally found a way to get my daughter to eat spinach. I made spinach parathas and told her they were 'Hulk parathas' that would make her strong. She ate THREE! Sometimes it's all about the marketing 😄
Has anyone else tried the superhero approach? Would love more ideas!
question to all the busy parents
Quick breakfast ideas for school mornings? I'm struggling to make healthy breakfasts on busy school mornings. My son has to leave by 7:30 AM and I work too. Currently giving him cereal which I know isn't the best.
What are some quick (under 10 minutes) healthy breakfast options? He's 8 years old and doesn't like eggs
meal prep time !
splashpage
NutriGuide
Personalized Meal Plans
Budget-Friendly Options
Get Started
login
NutriGuide
Create Account
Log In
Create Account
Email
Password
Get Started
Continue with Google
Continue with Apple
details
Help us understand your child
enter your child's name
your child's age
Phone number
Date of birth*
Select one...
select preferred language
Gender
Female
Male
Submit Form
onboard
Personalized recipe discovery
Next
Skip
mainpage
Recipe Corner
Learning Corner
Community Corner
Progress Overview
recipes
filter
Protien packed soya pulao
Veggie suji upma
community
COMMUNITY
success !!
question to all the busy parents

CONCLUSION

As first-year engineering students embarking on our social innovation journey, we recognize that **the lack of parental awareness about healthy eating for children** represents not just a nutrition problem, but a **multi-dimensional development challenge** that intersects with health, education, and equity.

Why This Problem Matters to Us:

Through our initial research and stakeholder mapping, we've discovered that this issue is deceptively complex. What appears as a simple "awareness gap" is actually rooted in economic constraints, cultural practices, time poverty, and systemic inequality in access to reliable nutrition information. Parents aren't failing their children - **the system is failing parents** by not providing accessible, practical, and culturally-relevant guidance.

Student Name with USN	Report		Se inar		Total (50)
	Content (15)	Organization of Report (10)	Presentation skills (15)	PPT (10)	
SANIDHYA RAI (0136/CS/2025-26)					

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